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Department of Computer Science

Exploratory Data Analysis

Data Engineering and Analytics

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Exploratory Data Analysis

- Classical statistics predominantely used inference (draw conclusion about large population based on small sample).
- John W. Tukey called for reformation in 1962
  - New scientific discipline “data analysis” given the available computing power.
  - 1977 book: “Exploratory Data Analysis” on the analysis of datasets.
  - Tukey also developed the fast fourier transform (Cooley-Tukey FFT) and the box plot.
  - Emphasized links between statistics, engineering and computer science (coined the terms software and bit).

Dimensions of Exploratory Data Analysis

Cross-classified along two dimensions:

(1) Graphical vs. non-graphical methods

- Graphical: diagrammatic summarization
- Non-graphical: summary statistics

(2) Univariate vs. Multivariate

- Univariate: analyze a single variable
- Multivariate: analyze two or more variables and explore their relationship (mostly bivariate analysis)

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Univariate Non-Graphical EDA: Overview

- Goals
  - Understand sample distribution
  - Make observations about population
- Tools
  - Tabulation for categorical data
  - Location (mean, median)
  - Spread (IQR, standard deviation, range)
  - Modality
  - Shape (skewness and Kurtosis)

Population vs. Sample

- Population: Entire set of observations we want to study.
  - Described by parameters, e.g., population mean μ, population standard deviation σ.
- Sample: Subset of the population used for analysis.
  - Described by sample statistics, e.g., sample mean $\bar{x}$, sample standard deviation s
- We use sample statistics to estimate unknown population parameters.
- How representative the sample is matters for generalizing results.
- Example: All students at a university (population) vs. 200 surveyed students (sample).
- You will see that for each of these concepts, there are slightly different methods.s

Tabulation for Categorical/Qualitative Data

What can be summarized for categorical/qualitative data?

- Range of values
- Frequency of values
- Relative frequency of values

	count	share
Drama	5076	0.243933
Comedy	3566	0.171368
Thriller	1664	0.079965
Romance	1644	0.079004
Action	1445	0.069441
Crime	1086	0.052189
Adventure	1003	0.048200
Horror	978	0.046999

Source: 04_dataset_analysis_exploratory.ipynb [Univariate Non-Graphical EDA]

Population Distribution for Quantitative Data

- Quantitative variable → analyze population distribution of variable based on the data of the observed variable
- Observed data is a sample (out of infinite number of possible samples)
- Characteristics of population distribution:
  - Location
  - Spread
  - Modality
  - Shape
  - Outliers

Location

- Variables may have thousands of distinct values.
- Estimate of location of the population distribution is a trivial form of summary.
  - What is a typical value for a variable?
  - Where is most of the data located?
  - What is the central tendency?

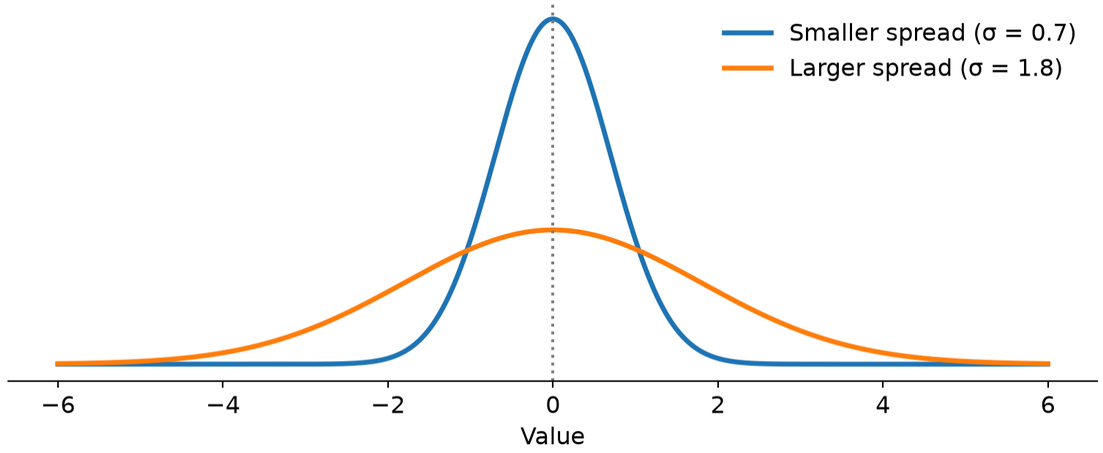
Estimates of Location

- Mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$
- Trimmed mean of sorted values (remove a certain percentage of extreme values from both ends before calculating the mean): $\bar{x}_k = \frac{1}{n-2k} \sum_{i=k+1}^{n-k} x_i$
- Weighted mean: $\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$
- Median: middle value of sorted data (or average of two middle values if even number of observations).
- Mode: most frequently occurring value.

Source: 04_dataset_analysis_exploratory.ipynb [Exploratory Data Analysis / Univariate Non-Graphical EDA]

Spread

- How far from the center are observations typically located?
- Two distributions may have the same mean but very different spread.
- Common measures:
  - Range
  - Variance and standard deviation
  - Interquartile range (IQR)

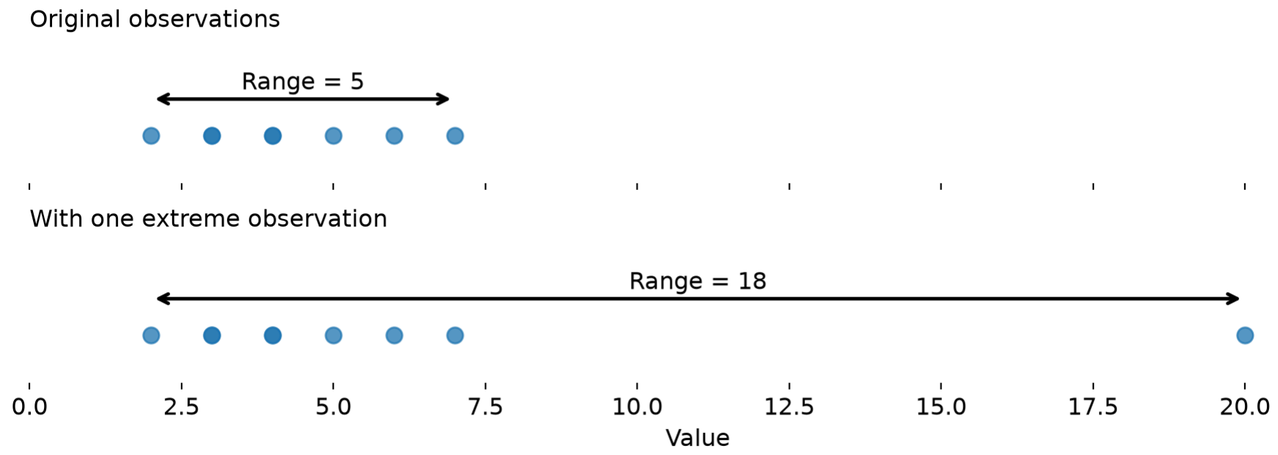


Spread: Range

- Range: difference between largest and smallest observation.

$$R = x_{\max} - x_{\min}$$

- Easy to calculate and interpret.
- Very sensitive to extreme observations.



Spread: Variance and Standard Deviation

- Population variance describes the spread of the entire population.

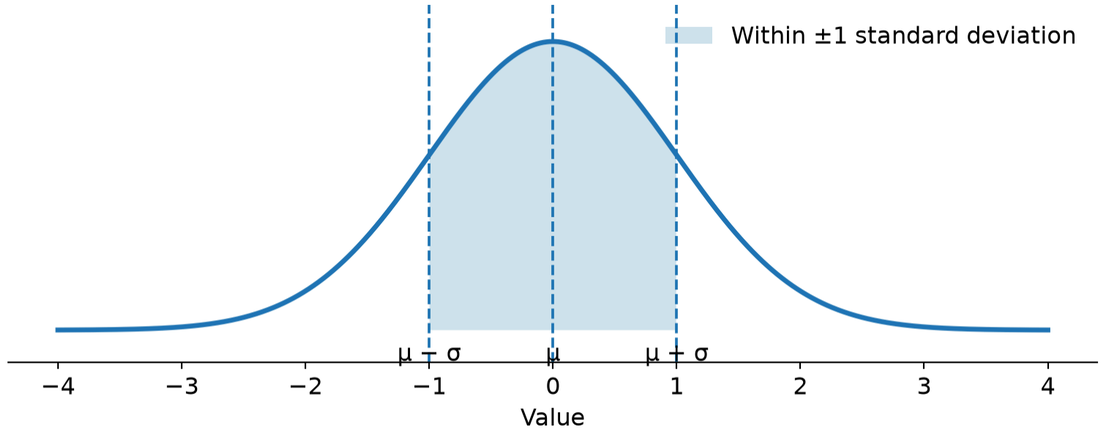
$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

- Sample variance estimates population variance using a sample.

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

- Why divide by $n-1$?
  - The population mean μ is unknown and estimated by $\bar{x}$.
  - Using $n-1$ corrects the systematic underestimation of population variance that results from dividing by n.
- Standard deviation is the square root of variance: $\sigma = \sqrt{\sigma^2}$ and $s = \sqrt{s^2}$.
- Standard deviation has the same unit as the observations.
- Both are sensitive to extreme observations.

Spread: Variance and Standard Deviation cont.

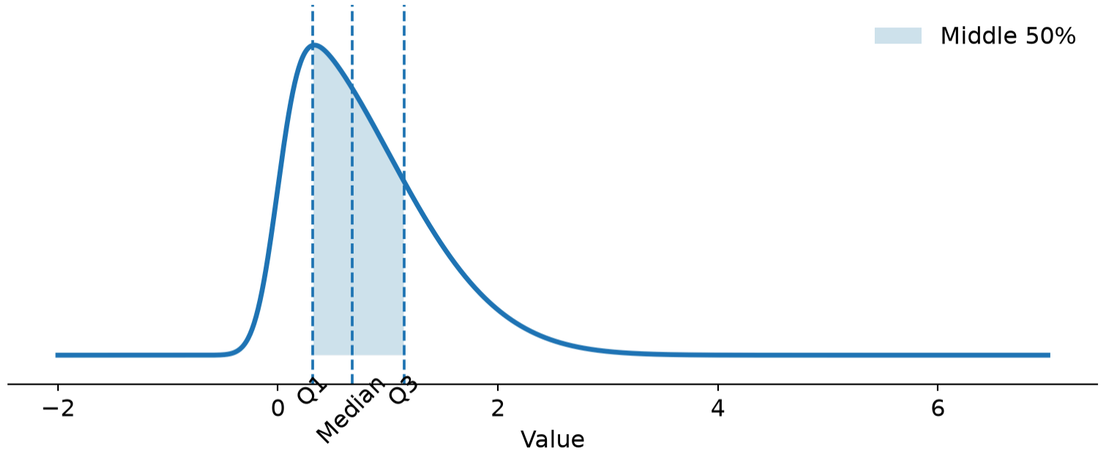


Spread: Interquartile Range

- Quartiles split the ordered observations into four parts.
- Interquartile range: width of the middle 50% of the data.

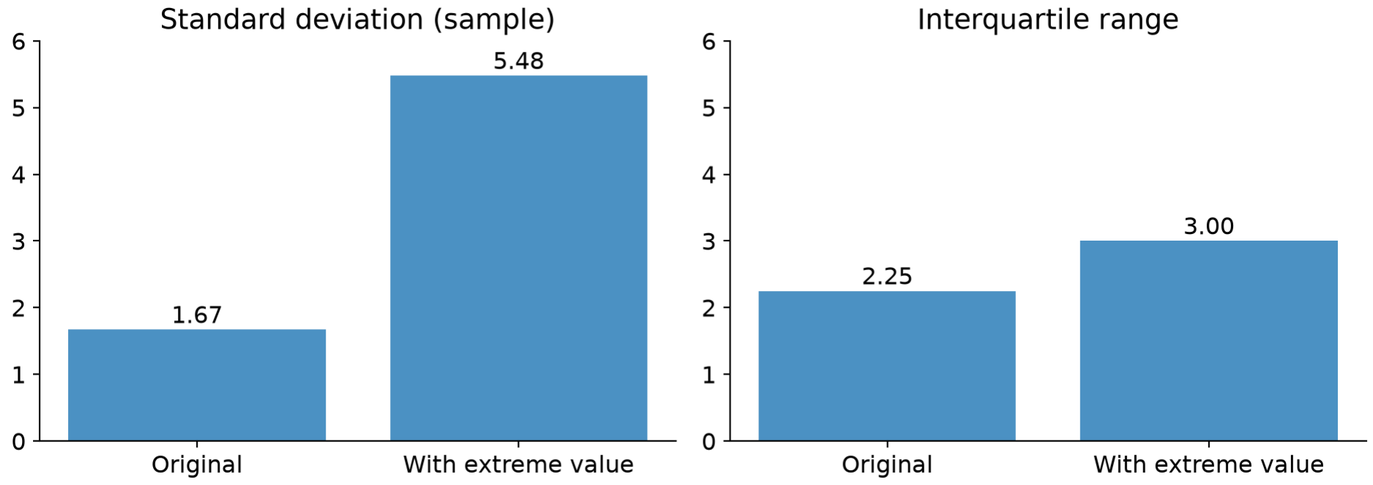
$$IQR = Q_3 - Q_1$$

- Less sensitive to extreme observations than variance or standard deviation.
- Particularly useful for skewed distributions.

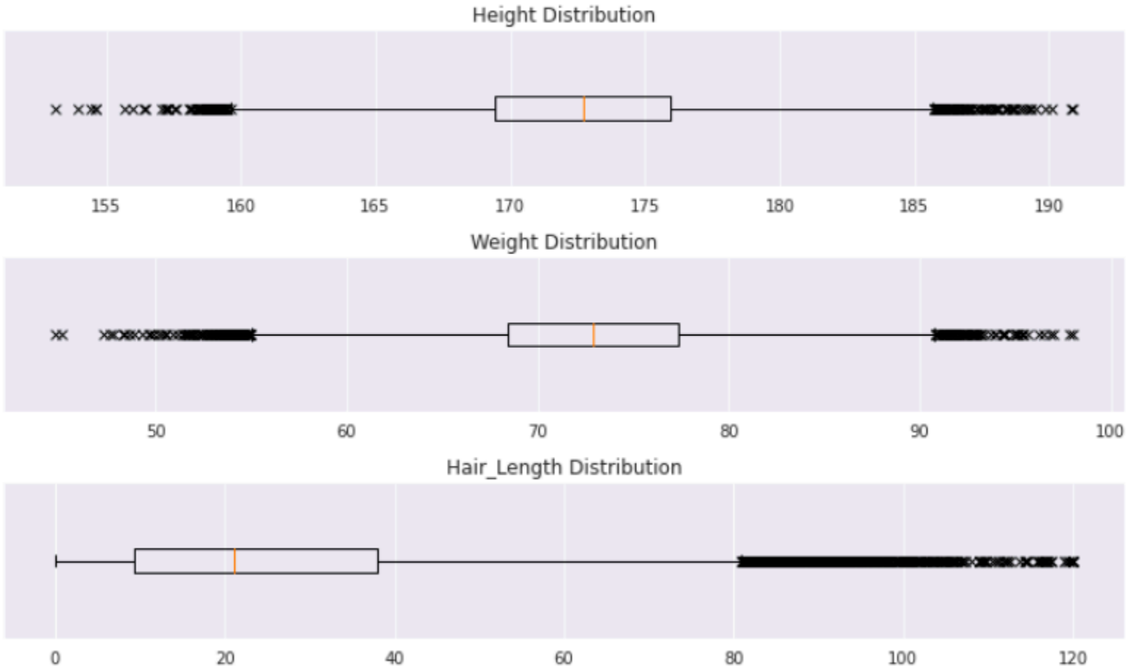


Spread: Robustness to Extreme Values

- An extreme observation can strongly affect range and standard deviation.
- Median and IQR are generally more robust.
- Choice of spread measure depends on the distribution and analysis goal.



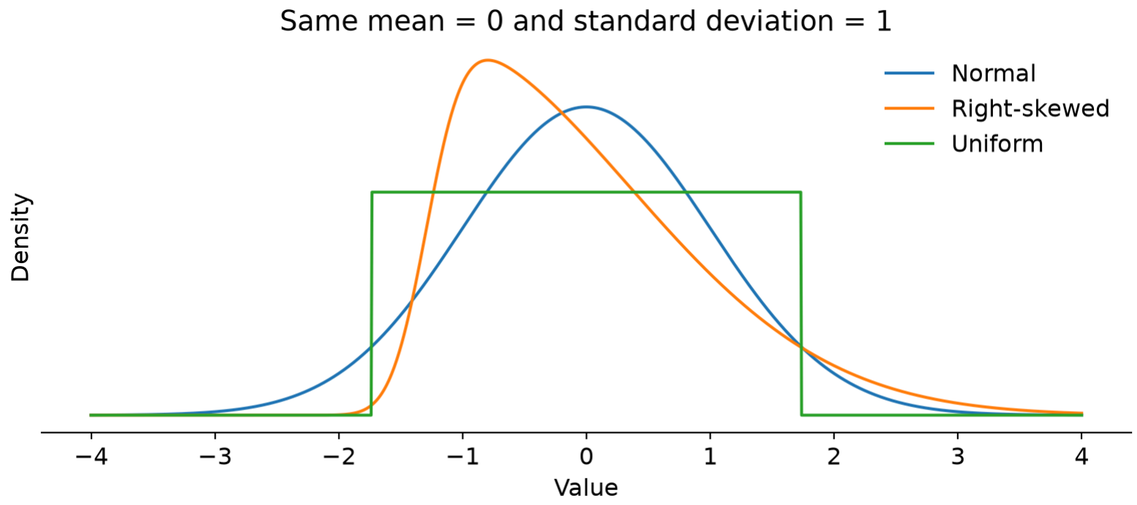
Spread: Robustness to Extreme Values cont.



Source: 03_data_analysis_cleaning.ipynb [Anomaly Detection / Outlier Detection / Interquartile Range]

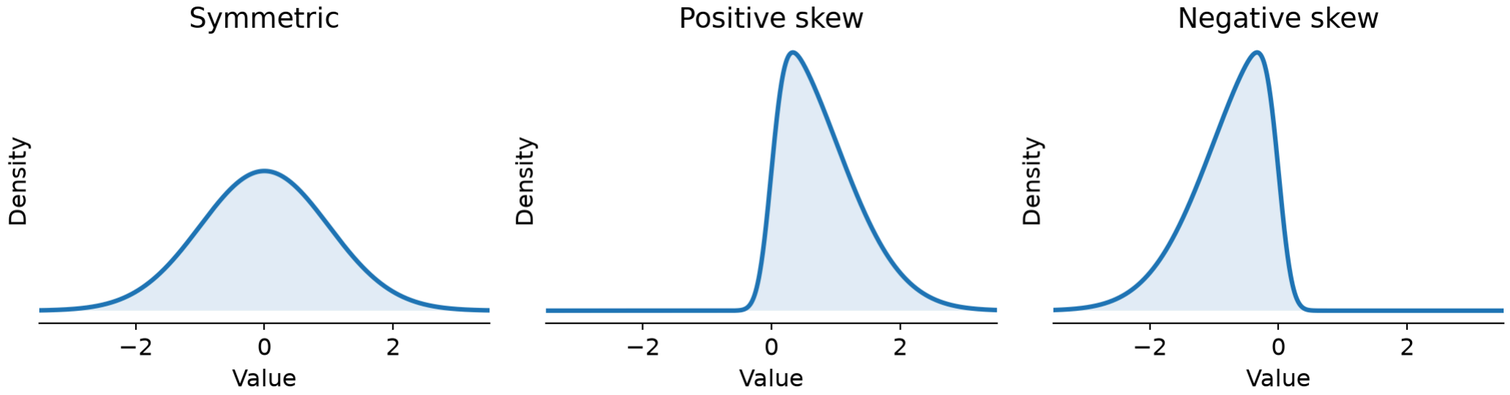
Shape

- Distributions with the same mean and standard deviation can still differ substantially in shape.
- Important characteristics:
  - Skewness: asymmetry of a distribution
  - Kurtosis: fourth standardized moment; sensitive to extreme deviations
  - Modality: number of peaks



Shape: Skewness

- Skewness describes the asymmetry of a distribution.
- Symmetric: matching shape on either side of the center.
- Positive skewness: longer or heavier right tail.
- Negative skewness: longer or heavier left tail.



Shape: Skewness (Mean and Median)

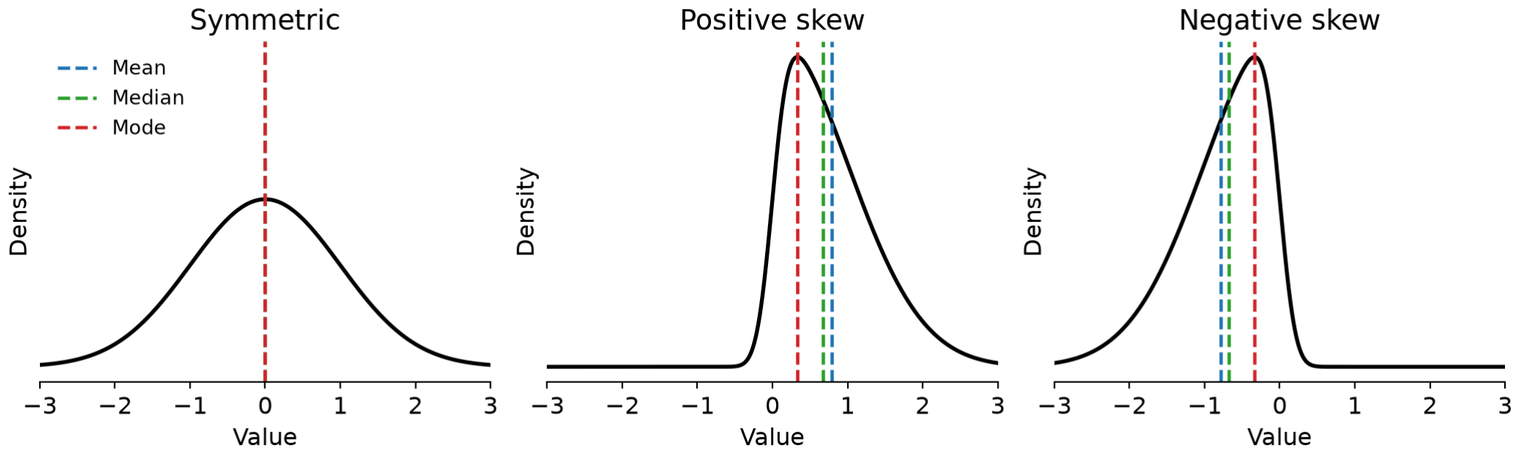
- A simple measure compares mean and median:

$$S = \frac{\mu - m}{\sigma}$$

  - μ: mean; m: median; σ: standard deviation
- $S > 0$: mean exceeds median; $S < 0$: mean is below median.
- Related to Pearson's second skewness coefficient (which uses a factor of 3).
- $S = 0$ does not necessarily imply symmetry.

Shape: Skewness (Location Measures)

- For many unimodal skewed distributions, the mean is pulled toward the longer tail.
- The familiar mean–median–mode ordering is an illustration, not a universal rule.



Shape: Skewness (Third Standardized Moment)

- Moment-based skewness:

$$\gamma_1 = \frac{\mu_3}{\sigma^3} = \frac{E[(X-\mu)^3]}{\sigma^3}$$

- $\gamma_1 > 0$: positive skewness; $\gamma_1 < 0$: negative skewness.
- Symmetric distributions have $\gamma_1 = 0$, if the moment exists.
- The converse is not necessarily true.
- Moment-based skewness is sensitive to extreme observations.

Shape: Kurtosis

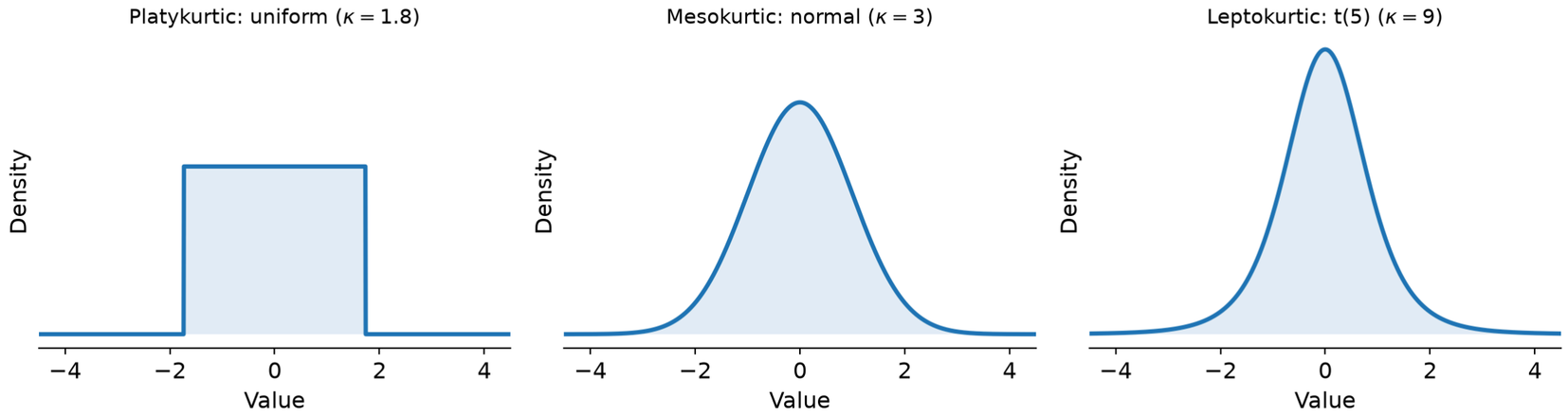
- Kurtosis is the fourth standardized moment:

$$\kappa = \frac{\mu_4}{\sigma^4} = \frac{E[(X - \mu)^4]}{\sigma^4}$$

- Particularly sensitive to extreme deviations from the mean.
- For a normal distribution, $\kappa = 3$.
- Excess kurtosis: $\gamma_2 = \kappa - 3$ (zero for a normal distribution).

Shape: Kurtosis (Interpretation)

- Mesokurtic: $\kappa = 3$ (normal-reference kurtosis).
- Leptokurtic: $\kappa > 3$.
- Platykurtic: $\kappa < 3$.
- Kurtosis is not simply the height of the central peak.



Shape: Kurtosis (Limitations)

- A few extreme values can substantially increase kurtosis.
- High kurtosis does not necessarily indicate erroneous observations.
- Low kurtosis does not imply that no outliers exist.
- Kurtosis alone cannot describe the entire distribution.
- Interpretation should be supported by graphical inspection.

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- **Univariate: analyze a single variable**
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Univariate Graphical EDA: Overview

- Goals
  - Complement non-graphical analysis
  - Analyze distribution of sample
- Means
  - Boxplots
  - Histograms
  - Bar Charts
  - Density Plots
  - Stem-and-leaf plots
  - Quantile-normal plots

Source: 04_dataset_analysis_exploratory.ipynb Univariate Graphical EDA

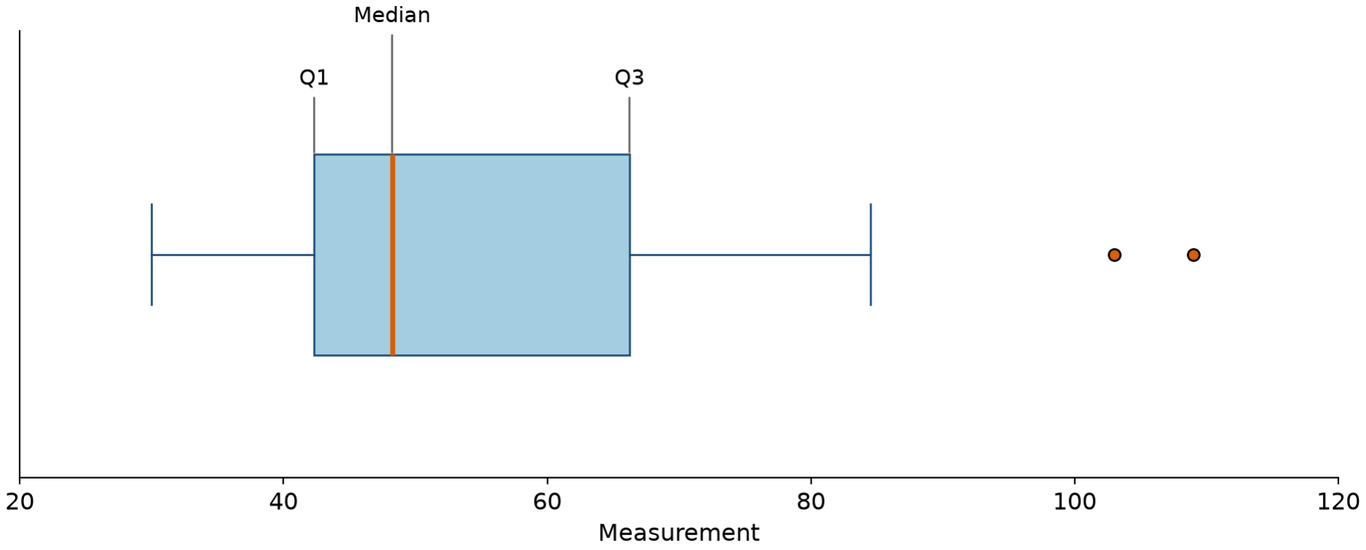
Univariate Graphical EDA

- Numerical variables: boxplots, histograms, density plots, violin plots, Q–Q plots
- Categorical variables: bar charts
- Complement numerical summaries of location, spread, and shape.
- Look for skewness, multiple modes, gaps, and potentially unusual observations.

> **Note**
>
> A plot is a representation of the data; choices such as bins, bandwidth, and axis scales affect what we see.

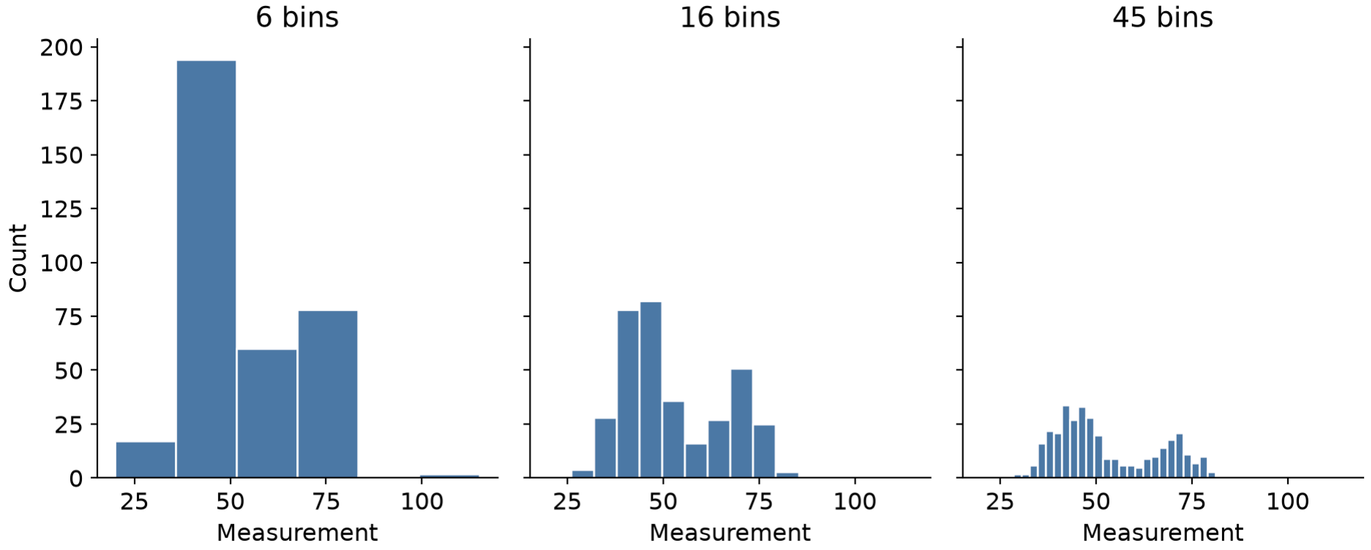
Boxplots

- The box spans the interquartile range (Q1 to Q3); the line marks the median.
- Standard Tukey whiskers extend to the most extreme observed values within 1.5 × IQR of the quartiles.
- Points beyond the whiskers are potential outliers, not necessarily errors.



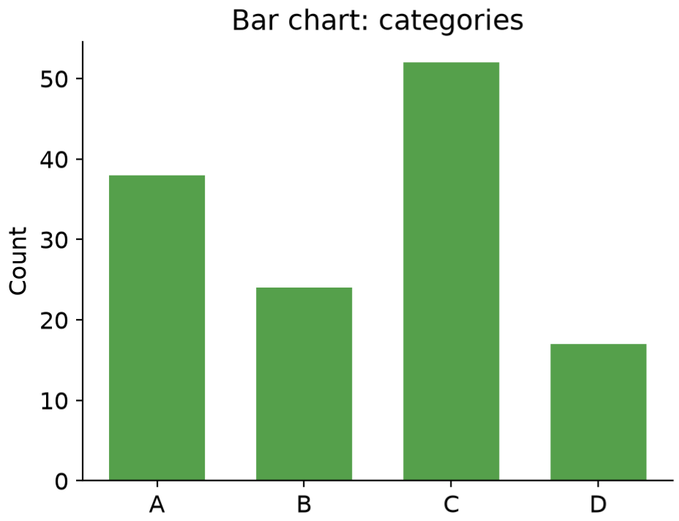
Histograms: Bin Width Matters

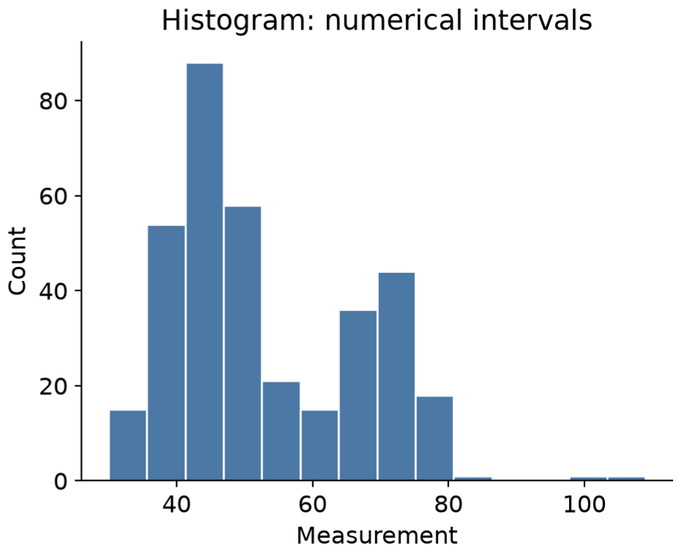
- Histograms group numerical values into intervals (bins).
- Bar height represents count (or density, depending on normalization).
- Different bin widths can emphasize or hide structure.



Bar Charts vs. Histograms

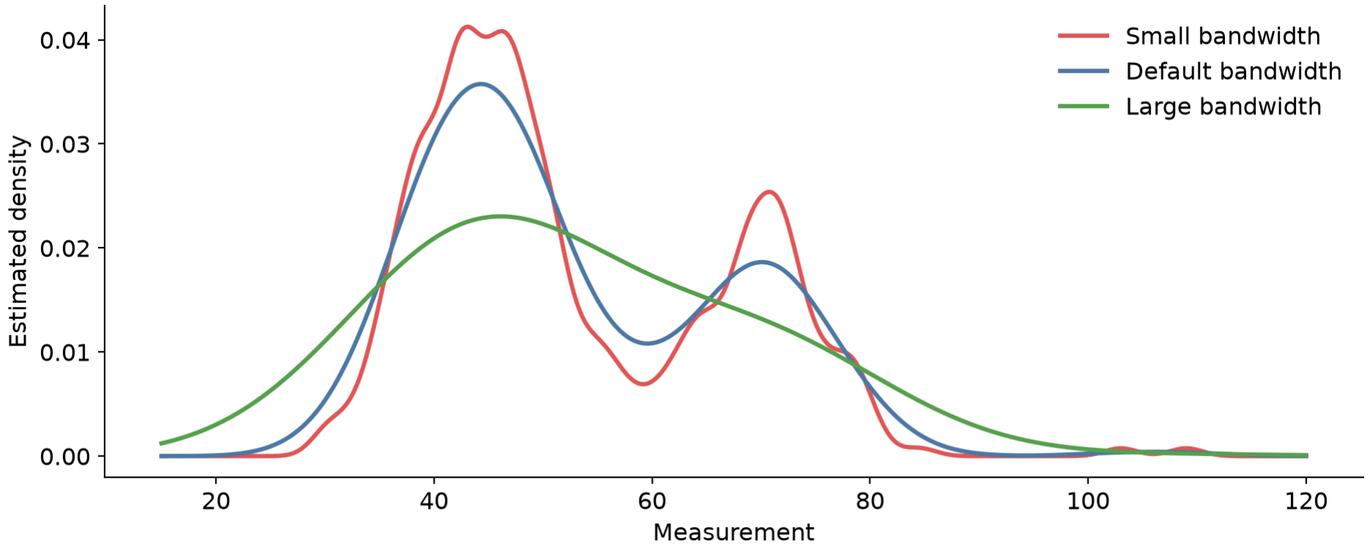
- Bar chart: categorical values; bars are separated and their order may be changed.
- Histogram: numerical intervals; bins follow a continuous numerical scale.
- Neither is interchangeable with the other.





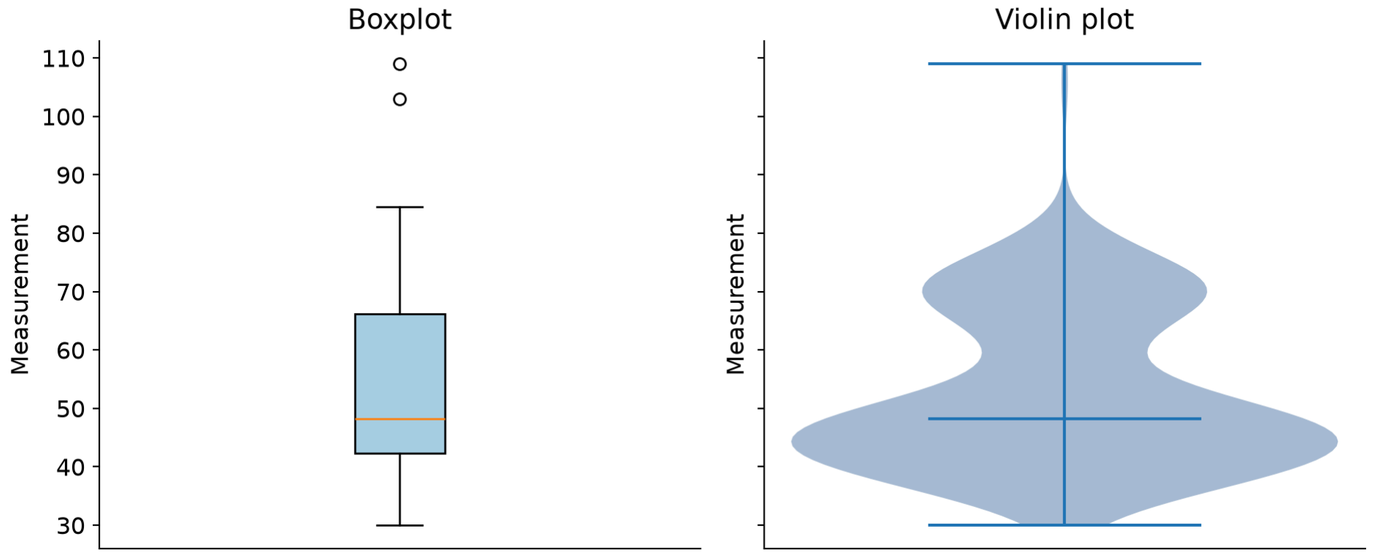
Density Plots: Kernel Density Estimation

- A kernel density estimate (KDE) is a smoothed estimate of a probability density.
- Bandwidth controls the amount of smoothing.
- Too little smoothing creates spurious bumps; too much can hide multiple modes.
- See (http://mathisonian.github.io/kde/) for an interactive demonstration.



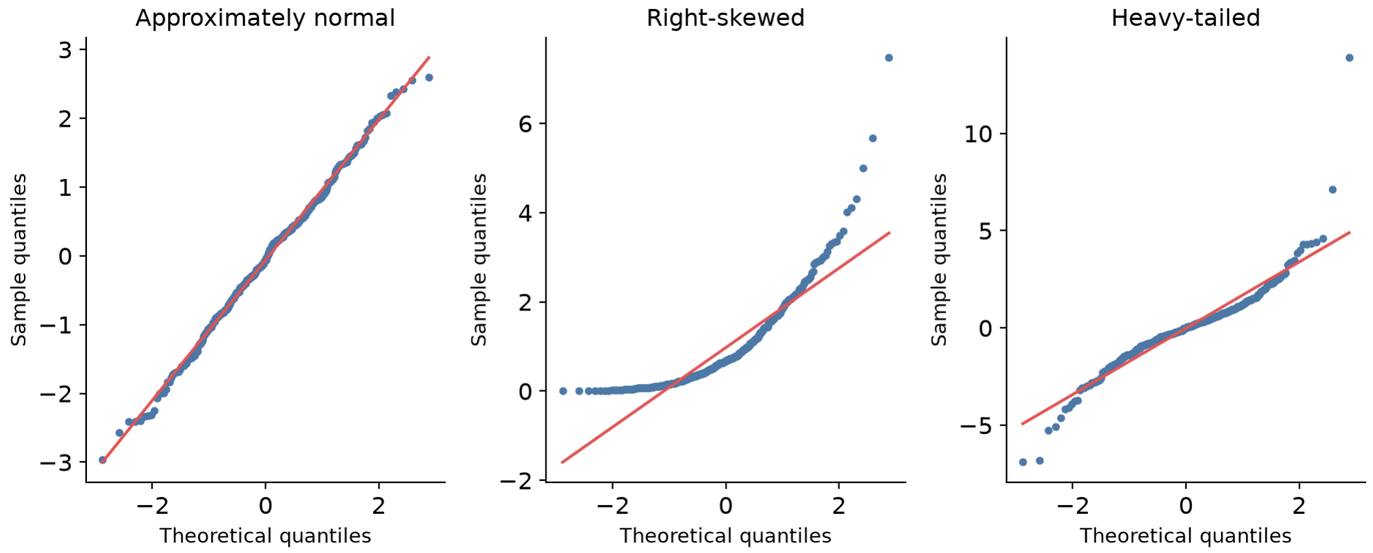
Violin Plots

- A violin plot displays a mirrored density estimate.
- It shows distribution shape; a boxplot more directly highlights quartiles and outliers.
- Violin shape depends on KDE smoothing and can extend beyond observed values unless constrained.



Normal Q–Q Plots

- Compare sample quantiles with quantiles from a theoretical normal distribution.
- Points approximately on a straight line suggest a similar distributional shape.
- Systematic curvature indicates skewness or differences in tail behavior.
- Process involves plotting the sample quantiles against the theoretical quantiles and assessing the linearity of the points.
- Theoretical quantiles come from the specified distribution (e.g., normal), while sample quantiles are computed from the observed data.



Choosing a Graphical EDA Method

Choosing the right visualization method depends on the type of data and the specific aspect of the data you want to explore.

- How are categories distributed? → Bar chart
- Where are the center, quartiles, and unusual values? → Boxplot
- What is the shape of a numerical distribution? → Histogram or density plot
- How do shape and quartiles compare? → Violin plot
- Does the sample resemble a normal distribution? → Normal Q–Q plot

ⓘ Combination of visualization methods

No single plot tells the whole story, combine multiple visualization methods to get a comprehensive understanding of the data and its underlying patterns. You might also add non-graphical summaries like tables or descriptive statistics.

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Multivariate Non-Graphical EDA: Overview

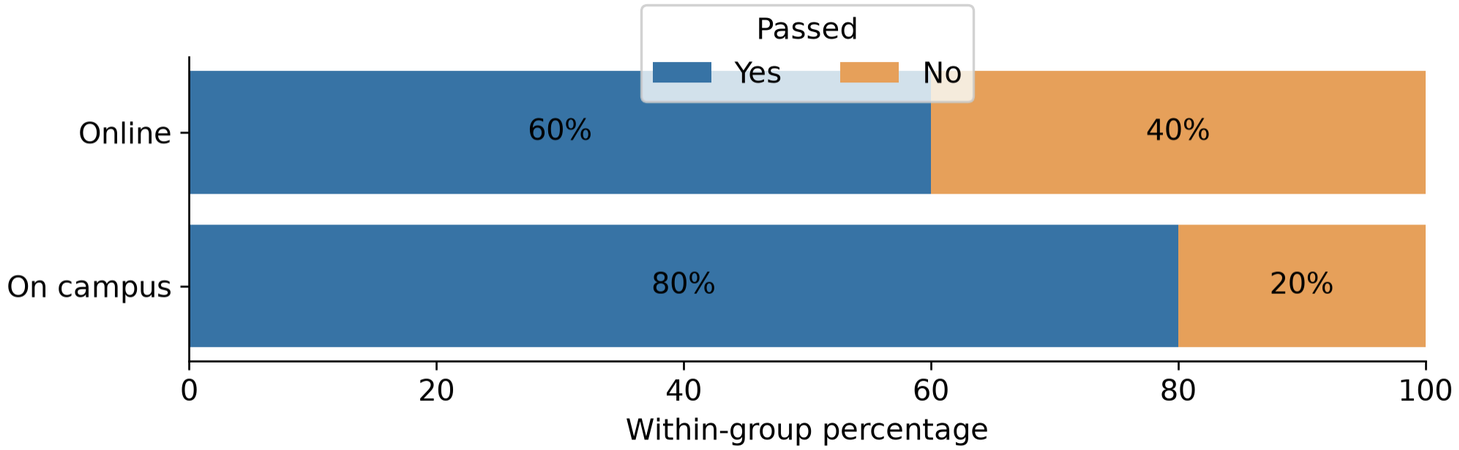
- Goal
  - Show relationships between two or more variables
- Tools
  - Cross-tabulation
  - Univariate statistics by category
  - Correlation and covariance
  - Correlation and covariance matrices

Multivariate Non-Graphical EDA

- Goal: Examine relationships between two or more variables.
- Common tools:
  - Cross-tabulation of categorical variables
  - Summary statistics by category
  - Covariance and correlation
  - Covariance and correlation matrices
- Numerical summaries should be complemented by graphical inspection.

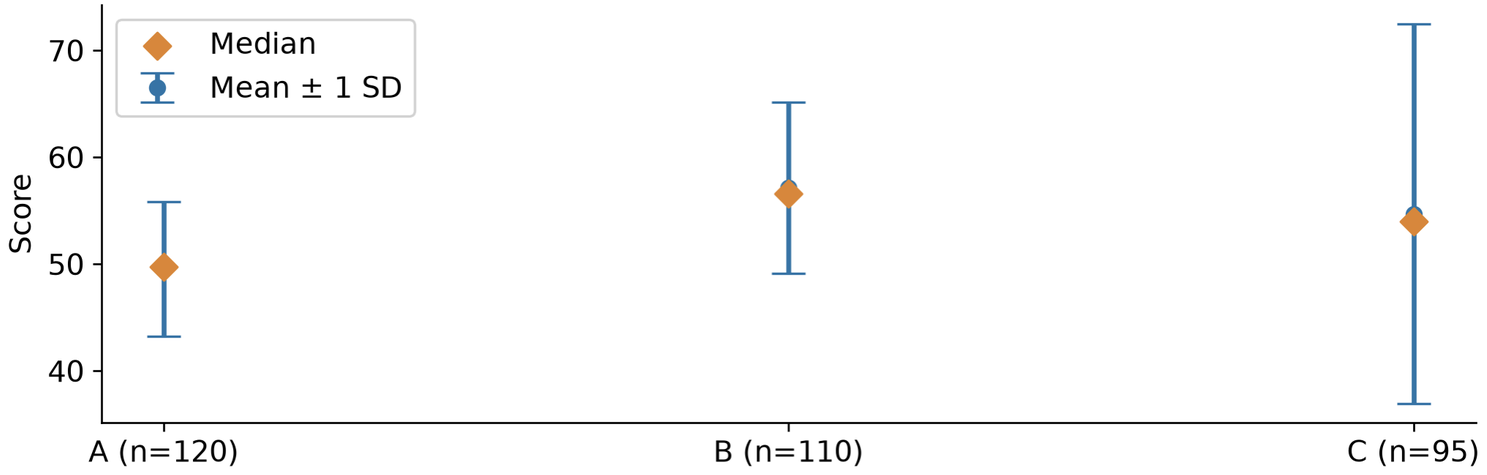
Cross-Tabulation

- A contingency table summarizes the joint frequencies of two categorical variables.
- Rows and columns represent categories; cells contain counts or proportions.
- Row percentages answer: How does the distribution of B differ across categories of A?



Univariate Statistics by Category

- For one categorical and one numerical variable, calculate summaries within each category.
- Compare location and spread, not only the means.
- Group sizes matter: a small group may have unstable estimates.



Covariance

- Covariance describes whether two numerical variables tend to deviate from their means in the same (opposite) direction.
- Population covariance:

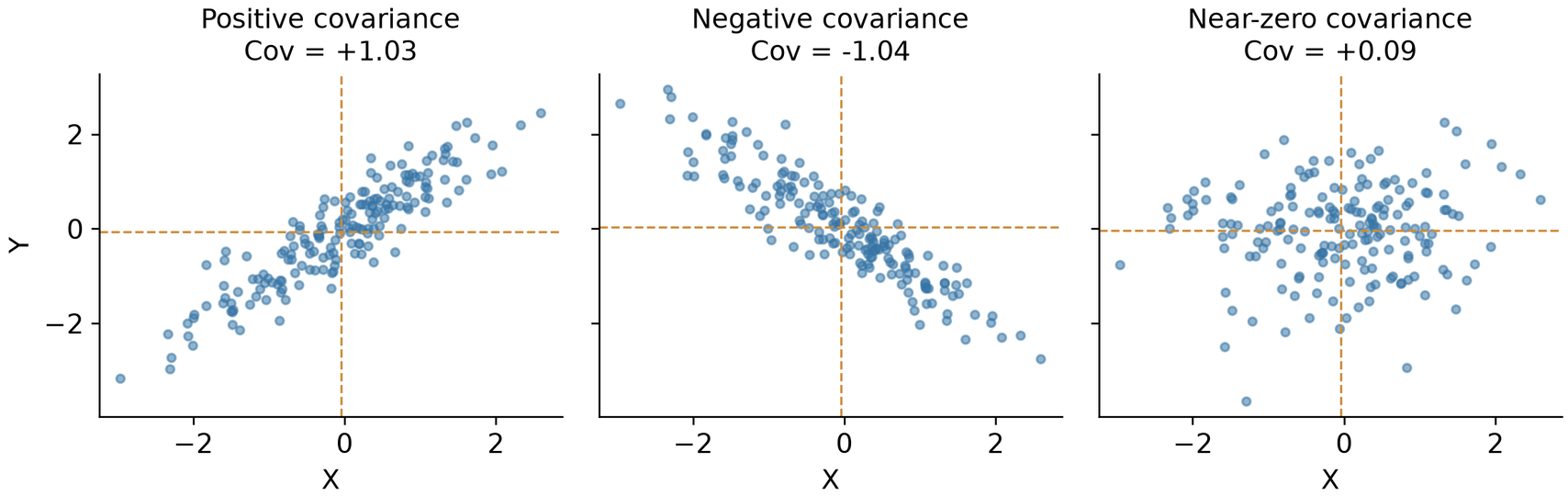
$$\operatorname{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

- Sample covariance:

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

- Positive: deviations often have the same sign.
- Negative: deviations often have opposite signs.
- Near zero: little *linear* co-variation; a nonlinear relationship may still exist.

Covariance: Visual Intuition



- Covariance depends on the measurement units and scale.
- Consequently, its magnitude is difficult to compare across variable pairs.

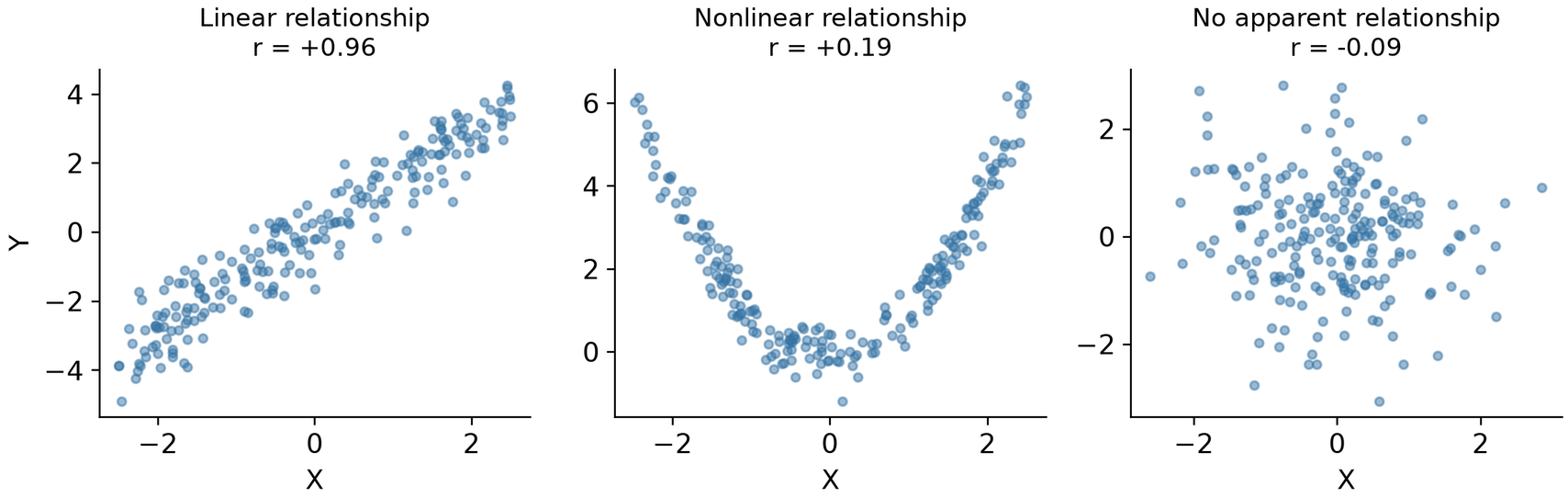
Pearson Correlation

- Pearson correlation is standardized covariance:

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- Sample version: $r = s_{XY} / (s_X s_Y)$.
- Values range from -1 to 1:
  - $r \approx 1$: strong positive linear relationship
  - $r \approx -1$: strong negative linear relationship
  - $r \approx 0$: weak or absent linear relationship
- Correlation is dimensionless, but sensitive to outliers.

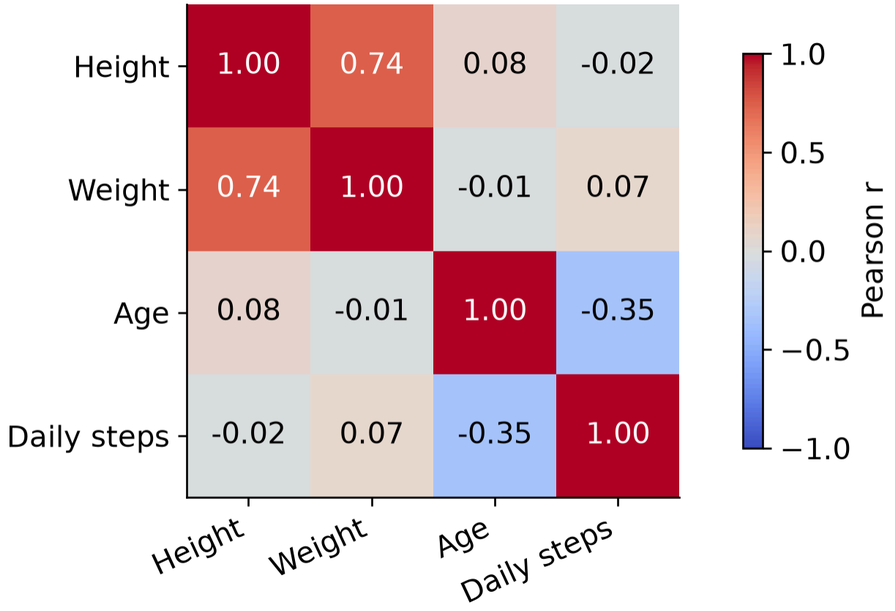
Pearson Correlation cont.



- Zero correlation does not imply independence.
- Correlation alone does not establish causation.

Covariance and Correlation Matrices

- For several numerical variables, compute all pairwise associations.
- A covariance matrix retains units and scales.
- A correlation matrix is standardized; its diagonal equals 1.
- A heatmap helps identify patterns, but inspect important pairs graphically.



Interpretation and Limitations

- Correlation-strength labels (e.g., “weak”, “strong”) are context-dependent.
- Always consider:
  - Sample size and uncertainty
  - Outliers and nonlinear relationships
  - Subgroups and possible confounding
  - Missing values and pairwise sample sizes
- Reporting correlation coefficients

π	**Interpretation/description**
0	No linear relationship
0.1-0.3	Weak linear relationship
0.3-0.5	Moderate linear relationship
0.5-0.7	Strong linear relationship
0.7-1.0	Very strong linear relationship

- Important: Numerical association measures summarize relationships; plots are needed to understand their structure.

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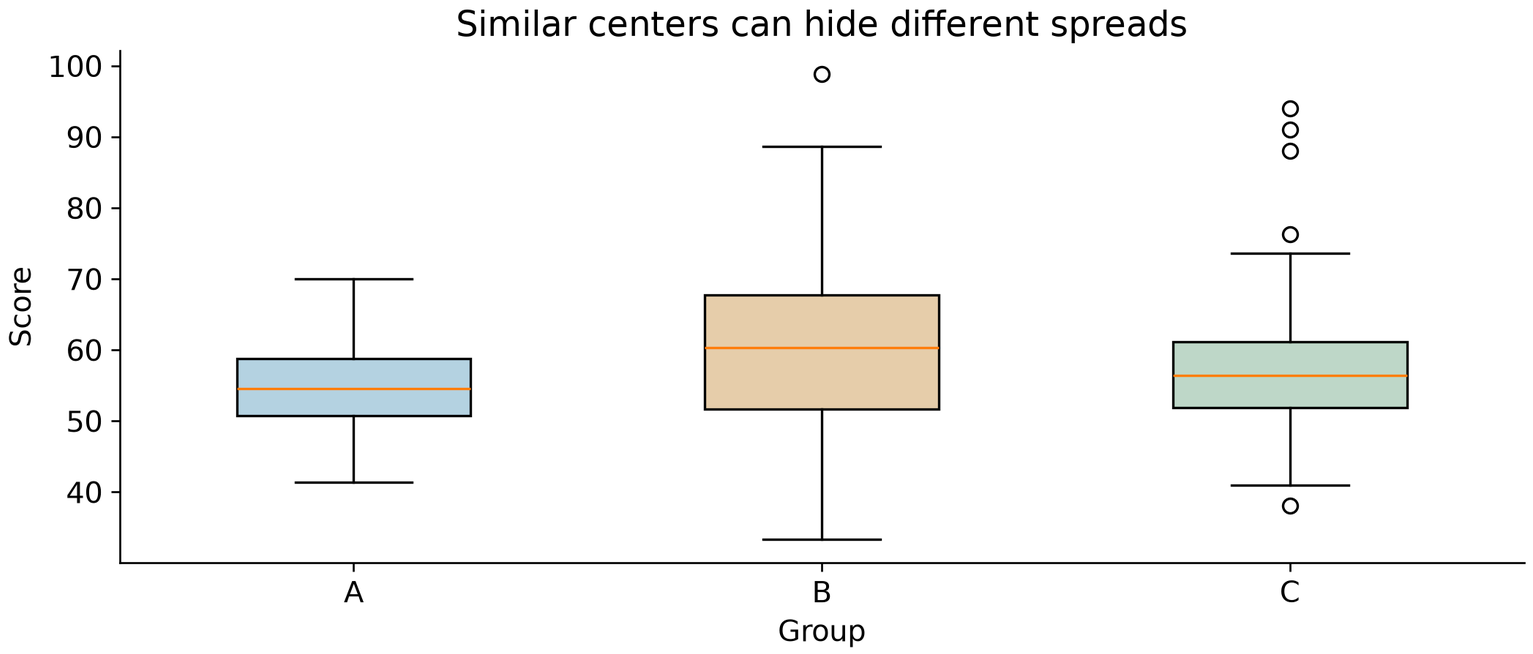
- Goals
  - Complement non-graphical analysis.
  - Analyze distribution of sample.
- Means
  - Univariate graphs by category (e.g., parallel box plots)
  - Scatterplots
  - Correlation

Multivariate Graphical EDA

- Goal: Explore relationships between two or more variables.
- Complement numerical summaries with visual inspection.
- Common techniques:
  - Grouped boxplots: categorical + numerical
  - Scatterplots: numerical + numerical
  - Scatterplot matrices: several numerical variables
- Look for trends, subgroups, nonlinear relationships, and unusual observations.

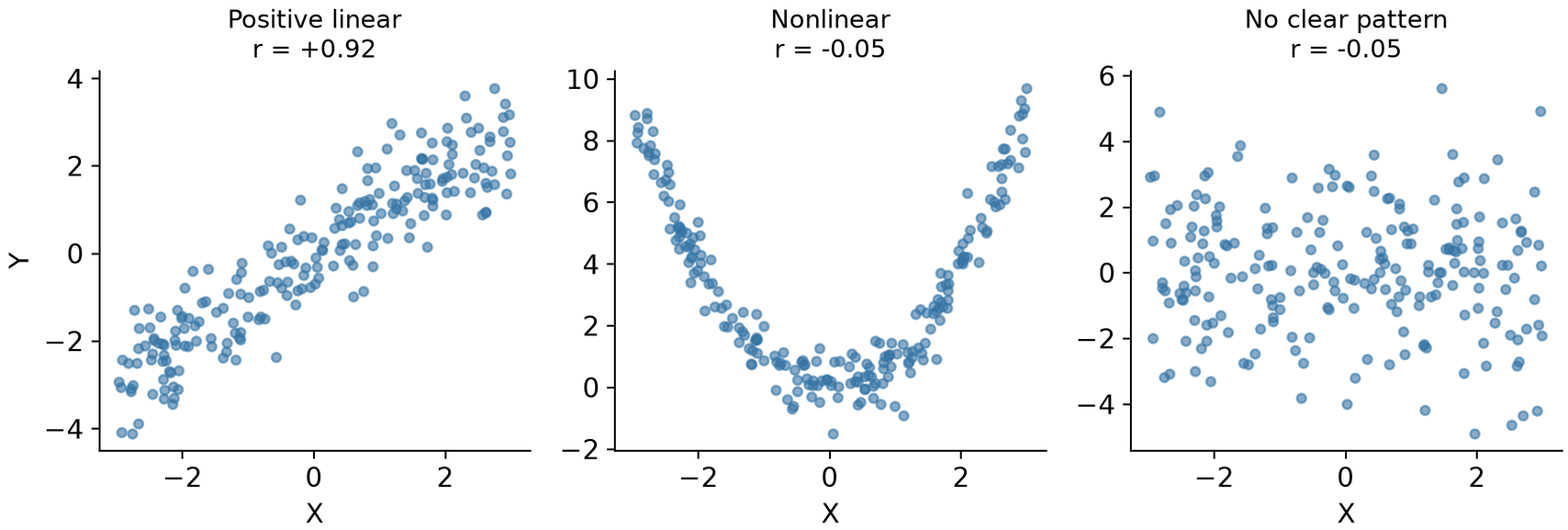
Univariate Graph per Category

- Compare the distribution of a numerical variable across categories.
- Grouped boxplots summarize median, quartiles, and potential outliers.
- Differences in spread may be as important as differences in location.



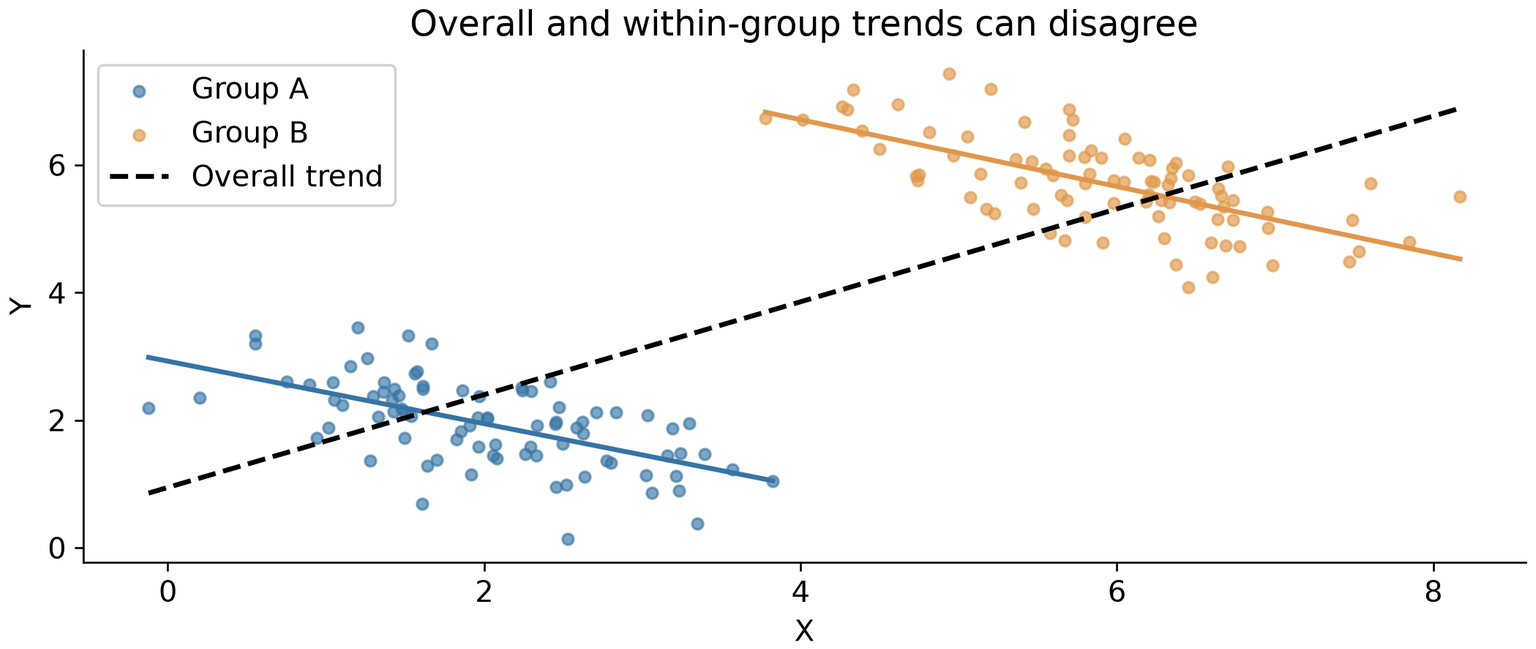
Scatterplots

- Show the pairwise relationship between two numerical variables.
- Each point represents an observation (x_i, y_i).
- Inspect direction, strength, nonlinear patterns, clusters, and outliers.
- A scatterplot can reveal relationships Pearson's r misses.



Scatterplots: Subgroups Matter

- The overall association may differ from associations within groups.
- Encode categories using colors or markers.
- Group structure may indicate confounding or aggregation effects.
- Correlation does not imply causation.



Scatterplot Matrix

- Examine pairwise relationships among several numerical variables.
- The diagonal shows the distribution of each variable.
- The off-diagonal shows pairwise scatterplots.
- Correlation matrices summarize linear relationships but cannot replace these plots.

